BY ORDER OF THE COMMANDER
AIR COMBAT COMMAND (ACC)

JOINT BASE LANGLEY-EUSTIS
INSTRUCTION 21-4004

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Maintenance

CRASHED, DAMAGED, OR DISABLED AIRCRAFT RECOVERY (CDDAR)

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This instruction implements the requirements and procedures for Crashed, Damaged, or Disabled Aircraft Recovery (CDDAR) as outlined in AFI 21-101, *Aerospace Equipment Maintenance Management*; AFI 21-101 CAFSUP, (*Aerospace Equipment Maintenance Management)*; AFMAN 32-4004, (*Emergency Response Operations), AFI 10-2501 (Air Force Emergency Management (EM) Program Planning and Operations)*, Base OPLAN 32-1, applicable 48- and 91- Series AFOSH Standards; and 1 FW Comprehensive Emergency Management Plan 10-2, TO 00-105E-9, *Aerospace Emergency Rescue and Mishap Response Information*, AFI 21-103, and aircraft specific Dash-2 and Dash-3 series TOs. Ensure that all records created as a result of processes prescribed in this publication are maintained in accordance with (IAW) AFMAN 33-363, Management of Records, and disposed of IAW the Air Force Records Disposition Schedule (RDS) located at **https://www.my.af.mil/gcss-af61a/afrims/afrims/rims.cfm**. Send comments, questions and suggested improvements to this instruction on AF Form 847, *Recommendation for Change of Publication*, through proper channels.

1. General. The 1st Maintenance Group (MXG) Commander is responsible for establishing a CDDAR program. The 1st Maintenance Squadron (1 MXS), Maintenance Flight will manage the program and has the primary responsibility for conducting CDDAR operations. 1 MXS Maintenance Flight will develop CDDAR procedures in coordination with the Base Fire Emergency Services (FES), Safety, Civil Engineering (CE), Readiness, Explosive Ordnance Disposal (EOD), Security, Bio-environmental Engineering (BEE), Airfield Manager and other on-/off-base agencies as applicable.

2. Assigned, Tenant, and Transient CDDAR Responsibilities.

2.1. First Responders will:

2.1.1. Deploy immediately to the disaster scene to provide initial Command and Control (C2) to prevent the loss of life, preserve valuable resources, protect the environment, and continue the mission. The basic CDDAR team (Crash Recovery Team [CRT]) is considered part of the First Responders and will follow the direction of the Incident Commander (IC).

2.1.2. All First Responders shall complete the Emergency Response Operations Course per AFI 10-2501.

2.2. Emergency Responders will:

2.2.1. Deploy after the First Responders to expand the C2 and provide additional support. The follow-on CDDAR team is considered part of the Emergency Responders and will follow the direction of the Fire Department. The CDDAR team will later become integral to the mishap aircraft removal.

2.2.2. All Emergency Responders shall complete the Emergency Response Operations Course per AFI 10-2501.

2.3. Emergency Operations Center (EOC), when activated as requested by the IC or EOC Director will:

2.3.1. Be comprised of Emergency Support Functions (ESF) as functional representatives and have varied responsibilities depending on resources, capabilities and mission of each installation. The EOC is usually comprised of an EOC Director, Manager and functional experts from CE, FES, EOD, Security Forces, Medical, BEE, Maintenance, Logistic Readiness Squadron (LRS), the Staff Judge Advocate (SJA), Force Support Squadron (FSS), Public Affairs (PA), Communications, Safety and Weather. Responding functional representatives perform duties inherent to their specific mission. If support requests exceed the capability of the installation, requests should be sent to higher headquarters.

2.3.2. Coordinate operations and support incident and base requirements with the IC, Installation Control Center (ICC), Recovery Operations Chief (ROC), Unit Control Centers, Command Post, specialized teams and civil and governmental authorities.

2.4. The Incident Commander (IC) will:

2.4.1. Direct actions to mitigate damage, save lives, restore primary mission assets and assist civil authorities.

2.4.2. Provide C2 of deployed elements by establishing required Air Force Incident Management System (AFIMS) sections, groups, divisions, etc.

2.4.3. Ensure controlled access to the accident site. Ensure personnel authorized access to the accident site is identified to Security Forces.

2.4.4. With the advice of the SJA, direct the establishment of National Defense Areas (NDA) and relay this information to the Command Post and responding members.

2.4.5. Direct the establishment of on-scene control and initial monitoring points.

2.4.6. Release information about the emergency response operation through PA.

2.4.7. Assess the threat of terrorists or potential protesters to response resources at the scene.

2.4.8. Work with mishap and accident investigation boards.

2.5. Recovery Operations Chief (ROC) will:

2.5.1. Receive status of operations from the IC and serve as the senior military representative until recovery operations are complete or until relieved by a higher authority or responsible agency. The ROC may become the IC when operations transition into a recovery phase.

2.5.2. Coordinate military activities with civil authorities as required.

2.5.3. Establish communication with the nearest military installation.

2.5.4. Coordinate required support for higher headquarters response elements deployed to the scene.

2.5.5. Work with mishap and accident investigation boards.

2.6. The 633d Civil Engineering Squadron (CES) representatives will:

2.6.1. Complete the following through the Asset Management Flight:

2.6.1.1. Provide environmental protection advice for compliance with local, state and national requirements.

2.6.2. Complete the following through Explosive Ordnance Disposal (EOD), (CES/CED):

2.6.2.1. Advise the IC on weapons recovery and supervise the initial “render-safe” procedures.

2.6.2.2. Provide support and technical guidance on explosive components and weapon component recovery and coordinate with Munitions Flight for additional support.

2.6.2.3. On initial entry of a nuclear weapons accident monitor for radiological hazards.

2.6.3. Complete the following through Fire Emergency Services (FES), (CES/CEF):

2.6.3.1. Establish and maintain C2 through the IC of the operations until incident transitions into a recovery phase. The IC also designates the Entry Control Point (ECP) location and determines the initial disaster cordon size.

2.6.3.2. Provide the hazard prediction capability operator with information concerning hazardous material type, source strength, amount, etc., to calculate a toxic corridor. 633 FES personnel may assume this responsibility, if the Emergency Management Flight is not immediately on site.

2.6.3.3. Order emergency withdrawal if necessary.

2.6.3.4. Ensure composite materials are extinguished and cooled to ambient temperature before the CRT can perform composite material containment, clean-up and disposal operations.

2.6.3.5. Brief the ROC upon arrival to the accident scene and turn over C2 when transition of incident goes to recovery.

2.6.4. Complete the following through the Operations Flight.

2.6.4.1. Assess damage to government and private real property.

2.6.4.2. Coordinate restoration and repair of property and provide other CE emergency support.

2.6.4.3. Coordinate and/or operate heavy equipment to provide access roads to the accident scene.

2.6.5. Complete the following through Readiness/Emergency Management Flight (CES/CEX):

2.6.5.1. Advise the IC/ROC concerning major accident and natural disaster response and recovery policies and procedures. Maintain communication link between the EOC, command post and accident site. Provide periodic situation updates at the scene.

2.6.5.2. Maintain a log of events for all on-scene actions and communications. Develop after-action reports and forward those reports to higher headquarters counterparts.

2.6.5.3. Perform hazard prediction using available software based on hazardous material type, source strength, amount involved, spill type, etc.

2.6.5.4. Advise on evacuation and cordon size.

2.6.5.5. Perform radiological surveys (radial or grid) under the direction of the Service Response Force.

2.6.5.6. Provide input to OPREP-3 reports through the installation EOC.

2.6.5.7. Provide and maintain Mobile Emergency Response Support Trailer.

2.6.5.8. Monitor air around the cordon to contain contamination within cordon.

2.7. The 633d Security Forces representative will:

2.7.1. Advise the IC/ROC on security measures and ensure that classified material is protected.

2.7.2. Establish and maintain physical security of the Disaster Cordon, NDA, ECP, On-Scene Control Point, Crime Scene and Parking/Staging Area as established by the IC.

2.7.3. Establish ECP procedures, controlling access to the accident site or disaster scene in coordination with the IC.

2.7.4. Coordinate with civilian law enforcement agencies.

2.7.5. In conjunction with the Staff JA, advise the IC/ROC on procedures for establishing a NDA.

2.8. The 633d Medical representative will:

2.8.1. Complete the following through the Emergency Department:

2.8.1.1. Coordinate with local medical facilities and direct the treatment and decontamination of medical casualties at those locations.

2.8.1.2. Advise the IC/ROC on blood-borne pathogen protection for emergency responders.

2.8.1.3. Act as a liaison with the base medical facility for on-base and off-base medical needs.

2.8.1.4. Provide medical support for responders and the accident investigation team as part of AFIMS.

2.8.1.5. Advise the IC/ROC on the status of medical treatment activities.

2.8.2. Complete the following through Bioenvironmental Engineering (BEE) Flight: (Health physicist or medical representative in the bioenvironmental engineer's absence) will:

2.8.2.1. Monitor personnel posted around the cordon to ensure they are not in a contaminated area.

2.8.2.2. Evaluate the occupational, radiological and environmental health hazards at or near the accident or disaster scene including presence of advanced aerospace materials, specifically composites.

2.8.2.3. Determine protective measures and equipment for personnel entering the accident or disaster scene.

2.8.2.4. Advise the IC/ROC when to evacuate personnel, including civilians.

2.8.2.5. Along with CE Readiness/Emergency Management Flight, determine the need for personnel monitoring, CCS procedures and contamination control requirements. In coordination with the IC/ROC, the BEE representative will support and provide advice on the processing of personnel out of the cordon through the CCS until it has been determined there is no threat of contamination.

2.8.2.6. Coordinate with IC and medical treatment representative regarding the monitoring and decontaminating of medical casualties and follow-on medical surveillance requirements for exposed first responders.

2.8.2.7. Deliberate with the mortuary affairs officer on procedures for decontaminating human remains.

2.9. The 633d Transportation representative will:

2.9.1. Advise the IC/ROC on the availability or limiting factors of transportation resources.

2.9.2. Obtain a cargo and/or passenger manifest from home station of airlift aircraft, when applicable.

2.9.3. Ensure the availability of transportation for EOC members (when required) from the assembly point to the designated ECP.

2.9.4. Provide the CRT with transportation resources as required (note: some heavy equipment may need to be contracted or sourced from another location).

2.10. The 633d Contracting Squadron (633 CONS) representative will:

2.10.1. Work closely with effected units to procure required supplies and equipment to satisfy any required CDDAR responsibilities.

2.10.2. Provide contracting support to fulfill any CDDAR requirements identified by the end-user. The end-user of any required items has the responsibility of ensuring the availability of all required items necessary to perform their CDDAR duties. For those items that have usability age limits and/or expiration dates, or otherwise do not lend themselves to emergency 'stockpiling,' a current source list for any necessary item should be maintained by the end-user in case of an emergency. 633 CONS, as part of their CDDAR responsibilities, will ensure a contingency contracting team is available on an emergency basis (at time of the mishap) to aid in implementing CDDAR procurement responsibilities. The satisfaction of any requirements identified to 633 CONS by the end-user can be satisfied through a variety of contractual instruments, to include purchase orders, contracts, delivery orders, calls against Blanket Purchase Agreements (BPAs), or the Government Purchase Card (GPC). In any case, 633 CPTS, in conjunction with the end-user, will need to determine the correct funds to use, and in the case of the GPC card, will need to load the funding onto the card prior to use. In addition, the end-user may use their organizational GPC to purchase any stockpiled items up front as part of their planning process.

2.11. The 633d Staff Judge Advocate (JA) will:

2.11.1. Provide legal advice on matters, to include claims, provisions for setting up an NDA, jurisdiction and coordinating Air Force personnel activity off-site.

2.11.2. Coordinate aircraft and missile accident investigation board activities.

2.12. The 633d Force Support representative will:

2.12.1. Coordinate food service and billeting requirements for deployed disaster response elements.

2.12.2. Provide search and recovery for human remains, disposition and other mortuary services.

2.13. The 633d Public Affairs (PA) representative will:

2.13.1. Ensure that public information concerning accidents and DOD personnel, equipment, property or other resources is released as applicable.

2.14. The 633d Safety representative will:

2.14.1. Monitor response activities for safety hazards and coordinate safety mishap investigation board activities.

2.15. The 1st Maintenance Squadron (MXS) will:

2.15.1. Provide Aerospace Ground Equipment (AGE) as needed for aircraft recovery to include:

2.15.1.1. NF-2/FL-1 Light cart (4 ea.)

2.15.1.2. MC-2A Low-Pack (2 ea.)

2.15.1.3. MC-7 Air Cart (1 ea.)

2.15.1.4. Air Bag Blower (2 ea.)

2.15.1.5. B-1 Maintenance Stand (2 ea.)

2.15.1.6. B-4 Maintenance Stand (2 ea.)

2.15.1.7. -60/-85 Generator Set (1 ea.)

2.15.1.8. Power Converter (F-22) (1 ea.)

2.15.1.9. Hydraulic Test Stand (1 ea.)

2.15.1.10. -10 Air Conditioner (1 ea.)

2.15.1.11. Polyalphaolefin (PAO) Coolant Cart (F-22) (1 ea.)

2.15.1.12. Aircraft Lifting Jacks (1 set)

2.15.1.13. MD-1 Aircraft Tow Bar (1 ea.)

2.15.1.14. Tank Dollies (2 ea.)

2.15.1.15. Heaters (2 ea.)

2.15.2. Provide Aircraft Structural Maintenance and Metals Technology qualified personnel as needed for aircraft recovery.

2.15.3. Provide qualified Munitions and Armament personnel as needed for rendering safe and recovering weapons.

2.15.4. Provide Egress Shop qualified personnel as required for ejection seat and canopy safing.

2.15.5. Provide Fuel Shop qualified personnel as needed for aircraft fuel problems and/or hydrazine response.

2.15.6. Provide Transient Alert assistance on transient CDDAR operations.

2.16. 1 MXS Maintenance Flight will:

2.16.1. Provide the CDDAR team chief who will be a SNCO or civilian equivalent (MXG/CC may waive grade requirement to TSgt or civilian equivalent), approved by the MXG/CC, and tracked on the Special Certification Roster (SCR) as required by AFI 21-101, Para 14.10.5.8.1.

2.16.2. Provide a basic CDDAR team for In-Flight Emergencies (IFEs) and Ground Emergencies (GEs). The basic CDDAR team will have at least three personnel consisting of at least one qualified CDDAR 7-level or above. The basic CDDAR team will consist of a tow super qualified CDDAR team supervisor, an additional CDDAR qualified member and a licensed tow vehicle operator. The owning unit may be tasked with providing an aircraft tow vehicle and a licensed tow vehicle operator as needed during IFEs and GEs. All CDDAR team members will be familiar with local hot brake and aircraft barrier engagement procedures as outlined in local checklists. The CDDAR team will not perform any recovery operations until cleared by the IC/ROC.

2.16.3. Dispatch the basic CDDAR team immediately if an emergency occurs during normal operating periods or duty hours. A basic CDDAR team will be on stand-by outside normal operating hours and will be recalled by the Maintenance Operations Center (MOC) in the event of an aircraft emergency. If the emergency requires more than the basic CDDAR team, the CDDAR team chief will recall additional CDDAR team members as needed. These members may be tasked to perform the duties outlined in the Attachment 2.

2.16.4. Upon notification, dispatch a CDDAR team chief to an off-base accident site to evaluate personnel, equipment and tools needed for CDDAR operations.

2.16.5. Coordinate with current crane rental contractor and contracting representative if the Grove RT990 crane is down for maintenance and one is needed for CDDAR operations.

2.16.6. Ensure the CDDAR team conducts CDDAR operations in a minimum time period consistent with the following considerations: requirement to open runway for operational use, prevention of secondary damage to aircraft and preservation of evidence for mishap or accident investigation board.

2.16.7. Tow all emergency aircraft clear of the active runway to allow the owning unit access to their aircraft and allow the CDDAR team to immediately regenerate. The owning unit will be responsible for completing the towing operation to the designated parking location.

2.16.8. In the event that a crashed, damaged, or disabled aircraft is on the runway the 1st FW/CC or designated representative will determine the degree of urgency required to clear the runway. If immediate aircraft removal priority is given, the ROC has the option of using heavy construction equipment for its removal. This action must be coordinated with Airfield Management. The ROC will direct the operation and assist as necessary to remove the aircraft from the runway as the situation warrants.

2.16.9. Conduct initial and annual CDDAR training in accordance with local course training documents. The course training documents are developed in conjunction with the Maintenance Training Flight (MTF).

2.16.10. Maintain, inspect, and store CDDAR equipment. Required inspections will be tracked and documented on the items applicable AF Form 1800, AFTO Form 244, or AFTO Form 95 (as required). Inspections not required to be loaded in TAS will only be loaded in TAS at the Maintenance Flight Chief's direction. This equipment should include, but is not limited to:

2.16.10.1. Crane (e.g., 20-ton, 50-ton, as applicable).

2.16.10.2. Aircraft Tow Vehicle (MB-2, MB-4 etc.)

2.16.10.3. 7.5 to 15 Ton Tractor

2.16.10.4. 25-50 Foot Flatbed Trailer

2.16.10.5. Composite Equipment Trailer

2.16.10.6. Aircraft Air Bags

2.16.10.7. Aircraft Lifting Slings

2.16.10.8. Belly Bands, Shackles and/or Spreader Bar

2.16.10.9. Aircraft Towing Cables

2.16.10.10. Disabled Wheel Dolly

2.16.10.11. Manual Steering Bar

2.16.11. Provide qualified personnel for containment, clean-up, and disposal of advanced aerospace materials/composites IAW T.O. 00-105E-9 and AFMAN 32-4004. These members will wear the proper Personal Protection Equipment (PPE). The minimum PPE requirement for burned composites will include hooded Tyvek suits, leather gloves, nitrile rubber gloves, steel toe boots and a HEPA full face respirator. (Members must be respirator qualified for the recovery, if there is a threat of contacting composite materials.)

2.16.12. Coordinate through IC/ROC for assistance from the Army in case of an aircraft crash requiring helicopter retrieval of wreckage.

2.16.13. Coordinate through IC/ROC for assistance from the Navy/Coast Guard in case of an aircraft crash involving water retrieval of wreckage.

2.17. The owning Aircraft Maintenance Unit (AMU) or hosting AMU will:

2.17.1. When needed, remove external fuel tanks, travel pods, special purpose pods, pylons, missiles, chaff, flare, etc.

2.17.2. Supply CDDAR team with equipment, vehicles and personnel during multiple or simultaneous emergencies or as needed for CDDAR operations.

2.17.3. Provide an aircraft tow vehicle and a licensed tow vehicle operator during IFEs and GEs, if needed.

2.18. The 1 OSS Weather representative will:

2.18.1. Advise the IC/ROC and CDDAR team on meteorological conditions that might affect operations.

2.19. The 1 MOS Maintenance Operations Center (MOC) will:

2.19.1. Coordinate CDDAR team requests with base agencies for assistance in CDDAR operations.

2.20. 1 MXG Quality Assurance (QA) will:

2.20.1. Provide the CDDAR team with aircraft weight and balance information as required.

2.21. The T-38 Adversary Air Program Managers will:

2.21.1. Provide a subject matter expert (SME) to advise CDDAR team during tenant and transient T-38 acft CDDAR operations.

2.21.2. Provide the CDDAR team with T-38 specific T.O.s, Job Guides, or Manuals.

2.21.3. Supply the equipment and personnel as needed for CDDAR operations.

2.21.4. Be responsible for the annual inspection of the T-38 Aircraft Tow Cable and Aircraft Lifting Sling and maintain equipment documentation.

2.22. The 158th FW Alert Detachment 1 will:

2.22.1. Provide a SME to advise CDDAR team during tenant and transient F-16 CDDAR operations.

2.22.2. Provide the CDDAR team with F-16 specific T.O.s, equipment and personnel as needed for CDDAR operations.

2.22.3. Inspect F-16 Aircraft Tow Cable and Aircraft Lifting Sling annually and maintain equipment documentation.

2.23. The 192d Air National Guard will:

2.23.1. Provide a basic CDDAR team (as identified in para 2.16.2.) for IFEs and GEs during Unit Training Assemblies (UTA).

2.24. NASA Langley will:

2.24.1. Provide specific T.O.s, equipment, and a SME to advise the CDDAR team during CDDAR operations of NASA owned aircraft.

KORVIN D. AUCH, Colonel, USAF
Commander, 633d Air Base Wing

ATTACHMENT 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 10-2501, *Air Force Emergency Management (EM) Program Planning and Operations*, 24 January 2007

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AFI 21-101, *Aircraft and Equipment Maintenance Management*, 26 July 2010

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AFMAN 33-363, *Management of Records*, 1 March 2008

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AFI 36-2251, *Management of Air Force Training Systems*, 5 June 2009

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CJCSM 3150.05C, 11 July 2007

LANGLEYAFBI 11-250, *Airfield Operations and Base Flying Procedures*, 23 February 2007

LANGLEYAFBI 48-137, *Respiratory Protection Program*, 30 August 2010

TO 00-105E-9, *Aircraft Emergency Rescue Mishap Response Information (Emergency Services),* 31 December 2008

LAFB Comprehensive Emergency Management Plan (CEMP) 10-2

1 MXG Operating Instruction 36-2232, *Aircraft Maintenance Training Program*, 30 October 2009

Abbreviations And Acronyms

AFI—Air Force Instruction

AFIMS—Air Force Incident Management System

AFMAN—Air Force Manual

AFOSH—Air Force Occupational Safety and Health

AFOSHSTD—Air Force Occupational Safety and Health Standards

AFTO—Air Force Technical Order

AGE—Aerospace Ground Equipment

BEE—Bio-Environmental Engineering

C2—Command and Control

CC—Commander

CCS—Contamination Control Station

CDDAR—Crashed, Damaged, or Disabled Aircraft Recovery

CE—Civil Engineer

CONS—Contracting Squadron

CTK—Composite Tool Kit

CRT—Crash Recovery Team

ECP—Entry Control Point

EOC—Emergency Operations Center

EOD—Explosive Ordnance Disposal

FES—Fire Emergency Services

FOD—Foreign Object Damage

GE—Ground Emergency

GPC—Government Purchase Card

IAW—In Accordance With

IFE—In-Flight Emergency

IMDS—Integrated Maintenance Data System

IC—Incident Commander

JA—Judge Advocate

JBLE—Joint Base Langley Eustis

MIL—Master Inventory List

MOC—Maintenance Operations Center

MXG—Maintenance Group

NASA—National Aeronautics and Space Administration

NDA—National Defense Area

OPR—Office of Primary Responsibility

OPREP—3-Operational Event/Incident Report

PMA—Portable Maintenance Aid

PPE—Personal Protective Equipment

PA—Public Affairs

ROC—Recovery Operation Chief

SJA—Staff Judge Advocate

SME—Subject Matter Expert

SNCO—Senior Non-Commissioned Officer

TBA—Training Business Area

ATTACHMENT 2

CDDAR TEAM POSITION RESPONSIBILITIES:

1. The Recovery Lead (CDDAR Team Chief) will:

1.1. Use this checklist to prepare the CDDAR Team for a safe and successful aircraft recovery operation.

1.2. Draw up a contingency plan and assign tasks to personnel in order to retrieve the damaged aircraft.

1.3. Ensure the entire recovery team is currently CDDAR qualified in their 623s/TBA and IMDS.

1.4. Contact the appropriate supporting agencies as needed.

1.5. Notify Airfield Operations of the boom extension length if the crane is being utilized.

1.6. Ensure all vehicles, AGE, and support equipment are available for the aircraft lift.

1.7. Ensure a fire bottle is available and serviceable if the FES is not at the scene.

1.8. Ensure aircraft is grounded and chocked if possible.

1.9. Ensure aircraft forms or PMA is on hand.

1.10. Ensure drivers' licenses and competency cards are on hand and up to date for all assigned drivers.

1.11. Conduct a safety briefing before the operation begins and prior to the aircraft removal.

1.12. Verify that the Composite Team's respirator fit tests are current.

2. The Composite Team Lead will:

2.1. Use this checklist to prepare the Composite Team for successful aircraft recovery operations.

2.2. Check out the Tuff Book from support and load the latest T.O. 00-105E-9 on it or download a copy from e-pubs.

2.3. Assign a trailer CTK Equipment Monitor and ensure all items in the trailer are inventoried and serviceable.

2.4. Ensure MIL and Master MIL match.

2.5. Check the entire trailer for possible foreign object damage (FOD) hazards/damage.

2.6. Ensure the trailer is secure and connected to the tow vehicle correctly.

2.7. Inspect, document, and start generator, if power is needed.

2.8. When assisting with donning and doffing of composite suits and respirators for the composite team, ensure all members receive medical evaluation (vital signs) by the Medical Group/FES before and after entry into the contaminated area.

2.9. Properly dispose of contaminated consumables and replace them as soon as possible.

3. The Crane Operator will:

3.1. Follow this checklist to prepare the crane for a successful operation.

3.2. Ensure his/her driver's license and competency card are on person and up to date.

3.3. Ensure the vehicle's 1800 is signed off (which also covers the fire extinguisher).

3.4. Check the entire crane for possible FOD hazards/damage.

3.5. Ensure all CTK items are present and serviceable.

3.6. Wear proper hearing protection as required.

3.7. Know hand signals, correct switch operation, and crane limitations.

4. The Ground Lead will:

4.1. Use this checklist to prepare the CDDAR Team for a successful aircraft recovery operation.

4.2. Ensure the current PMA/TOD, Technical Orders, or applicable Job Guides are available and being utilized.

4.3. Ensure there is a serviceable fire bottle available.

4.4. Ensure all AGE equipment is signed off on the AFTO Form 244.

4.5. Ensure personnel are operating powered and non-powered AGE correctly.

4.6. Ensure everyone performs their assigned tasks safely and correctly.

5. The Aircraft Top Lead will:

5.1. Use this checklist to prepare the CDDAR Team for a successful aircraft recovery operation.

5.2. Ensure everyone is working in their assigned area.

5.3. Ensure current Technical Orders, TOD, or Job Guides are being utilized.

5.4. Ensure the connection of the sling is done safely and correctly.
5.5. Utilize tool bags as needed and place tool boxes (if used) on protective matting.

6. The Tractor Trailer Operator will:

6.1. Use this checklist to prepare the 7.5 Ton Tractor and 40 ft Trailer for a successful operation.

6.2. Ensure his/her driver's license and competency card are on person and up to date.

6.3. Sign off both 7.5 Ton Tractor and 40 ft Trailer 1800s.

6.4. Check the entire tractor trailer for possible FOD hazards/damage.

6.5. Use spotter for backing and movement in congested areas.

6.6. Allow the appointed CTK Equipment Monitor to complete their required checks.

7. The CTK/Equipment Monitor will:

7.1. Use this checklist to ensure all tools and equipment are accounted for and serviceable.

7.2. Sign out the Crash Trailer keys from the support section.

7.3. Ensure MIL and Master MIL match.

7.4. Inspect, document, and start generator if power is needed.

7.5. Inventory all tools and equipment and notify CDDAR Team Chief of shortages.

7.6. Check the entire trailer for possible FOD hazards/damage.

7.7. Inspect equipment, Form 95s (if applicable), load certification (if applicable) and Form 244s.

7.8. Sign off required Form 244 inspections prior to the equipment leaving the trailer.

7.9. Document all tools and equipment leaving the trailer.

8. The Air Bag Cell Lead will:

8.1. Use this checklist to ensure their air bag position is prepared for a successful aircraft recovery operation.

8.2. Ensure Technical Orders, TOD, or Job Guides are being utilized.

8.3. Ensure everyone is working their assigned area and performs their assigned duties safely and correctly.

8.4. Ensure all powered and non-powered AGE is signed off and used correctly.

8.5. Level the ground and/or position dunnage as required.